Fundamentals of Linear Control:
A Concise Approach

Maurício C. de Oliveira

Chapter 4: Feedback Analysis

linearcontrol.info/fundamentals

Recall from Chapter 3

Time-invariant linear system in the time-domain

Impulse response

$$g(t), \quad t \geq 0$$

Convolution

$$y(t) = \int_{0^-}^{\infty} g(t-\tau)\, u(\tau)\, d\tau$$

Time-invariant linear system in the frequency-domain

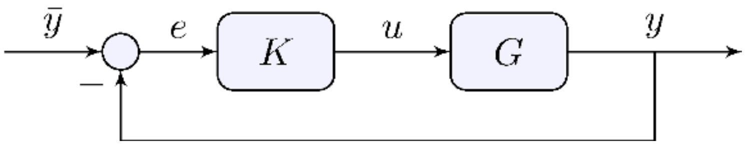
Transfer-function

$$G(s) = \mathcal{L}\{g(t)\}$$

Product

$$Y(s) = G(s)\, U(s), \qquad Y(s) = \mathcal{L}\{y(t)\}, \qquad U(s) = \mathcal{L}\{u(t)\}$$

4.1 Tracking, Sensitivity, and Integral Control



Loop relations

$$Y(s) = G(s)\,U(s), \qquad U(s) = K(s)\,E(s), \qquad E(s) = \bar{Y}(s) - Y(s)$$

Closed-loop sensitivity transfer-function

$$E(s) = S(s)\,\bar{Y}(s), \qquad S(s) = \frac{1}{1 + G(s)\,K(s)}$$

Short-hand notation

$$e = S\,\bar{y}, \qquad S = \frac{1}{1 + G\,K}, \qquad \text{etc}$$

Example: car model with proportional feedback

Open-loop transfer-function

$$G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}}$$

Closed-loop sensitivity transfer-functions

$$S(s) = \frac{1}{1 + K\frac{\frac{p}{m}}{s+\frac{b}{m}}} = \frac{s + \frac{b}{m}}{s + \frac{b}{m} + K\frac{p}{m}}$$

Closed-loop stability

$$-\left(\frac{b}{m} + \frac{p}{m}K\right) < 0 \qquad \Longrightarrow \qquad K > -\frac{b}{p}$$

Tracking error

Reference input

$$\bar{y}(t) = \bar{y}\cos(\omega t + \phi), \quad t \geq 0$$

Steady-state from frequency response

$$e_{\mathrm{ss}}(t) = \bar{y}\,|S(j\omega)|\cos(\omega t + \phi + \angle S(j\omega)) \qquad \Longrightarrow \qquad |e_{\mathrm{ss}}(t)| \leq |S(j\omega)|\,|\bar{y}|$$

e.g. car model with a step reference

$$|e_{\mathrm{ss}}(t)| \leq |S(0)|\,|\bar{y}| = \frac{\frac{b}{m}}{\frac{b}{m} + \frac{p}{m}K}|\bar{y}| = \frac{1}{1 + \frac{p}{b}K}|\bar{y}|$$

Asymptotic tracking

Definition

Error converges to zero for large time

$$\lim_{t \to \infty} e(t) = 0$$

Stability of S is not enough. It is necessary that

$$E = S\,\bar{Y}$$

Poles of $\bar{Y}$ must be cancelled by zeros of S!

Lemma 4.1

Let** $S = (1 + G\,K)^{-1}$ **be asymptotically stable and

$$\bar{y}(t) = \bar{y}\cos(\omega t + \phi)$$

***If** $G\,K$ **has a pole at** $s = j\omega$ **then** $\lim_{t \to \infty} e(t) = 0$*

Proof: S has a zero at $s = j\omega$

Example: toilet bowl model

Open-loop transfer-function

$$G(s) = \frac{1}{sA}$$

Closed-loop sensitivity transfer-function

$$S(s) = \frac{1}{1 + G(s)\,K} = \frac{1}{1 + \frac{K}{sA}} = \frac{s}{s + \frac{K}{A}}$$

Asymptotically stable for $K > 0$

S has a zero at zero: asymptotic tracking of step (constant) reference inputs

Example: car model with integral feedback

Open-loop transfer-function

$$G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}}$$

Integral control

$$K(s) = \frac{K}{s}$$

Closed-loop sensitivity transfer-functions

$$S(s) = \frac{1}{1 + \frac{K}{s}\frac{\frac{p}{m}}{s+\frac{b}{m}}} = \frac{s\left(s + \frac{b}{m}\right)}{s^2 + \frac{b}{m}s + K\frac{p}{m}}$$

Asymptotically stable for $K > 0$

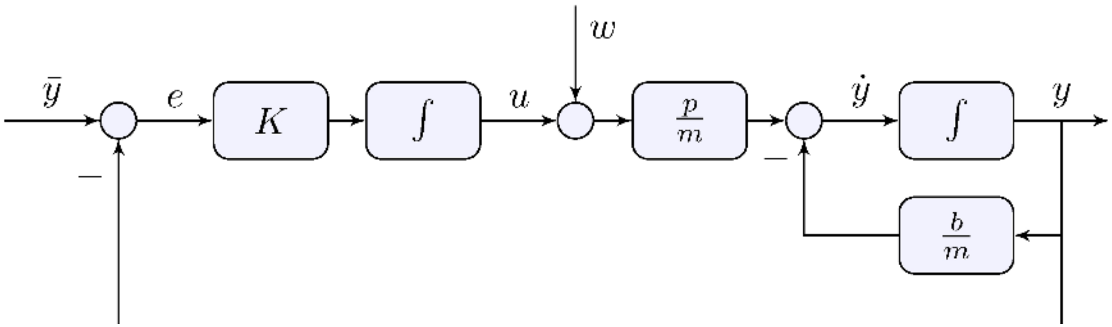
S has a zero at zero: asymptotic tracking of step (constant) reference inputs

Example: car model with integral feedback

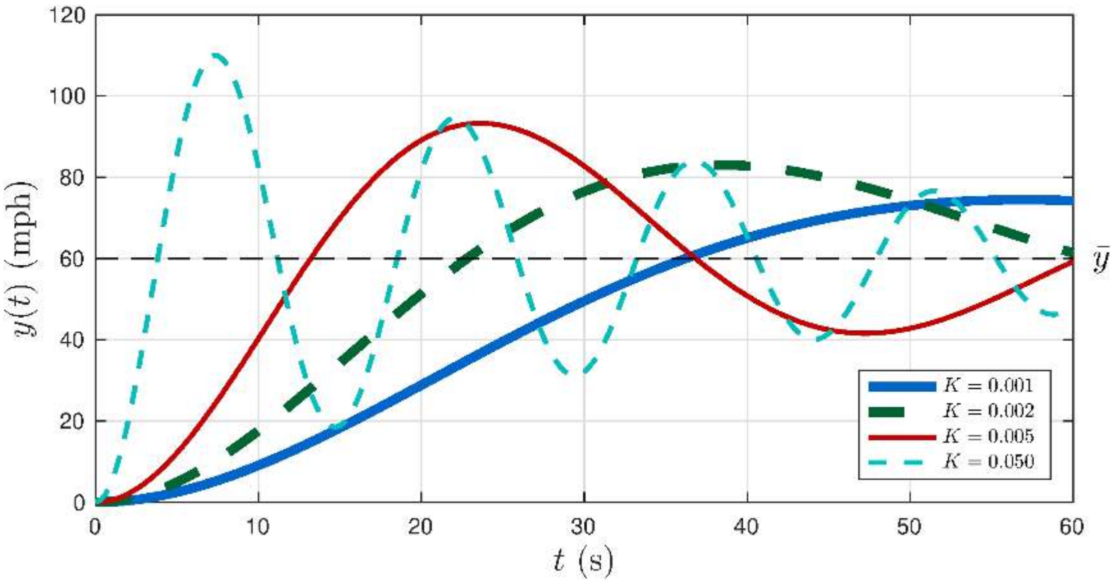
Controller

$$U(s) = \frac{K}{s} E(s) \qquad \Longrightarrow \qquad u(t) = u(0) + K \int_0^t e(\tau)\, d\tau$$

System + Controller block-diagram



Example: car model with integral feedback



Transient response has oscilations!

4.2 Stability and Transient Response

Some facts

- The zeros of $1 + GK$ are the poles of S
- The poles of the product $G\,K$ are the zeros of S
- When G and K are rational the order of S is the order of the product $G\,K$

Example: rational transfer-function with no pole-zero cancellations

$$G = \frac{N_G}{D_G}, \qquad K = \frac{N_K}{D_K}, \qquad S = \frac{1}{1 + \frac{N_G}{D_G}\frac{N_K}{D_K}} = \frac{D_G D_K}{D_G D_K + N_G N_K}$$

For tracking

- Add poles to K so they become zeros of S
- More complex K leads to more complex S
- Watch out for stability and oscillations, e.g. car model with integral feedback

Example: car model with proportional plus integral feedback

Proportional plus integral control

$$G(s) = \frac{p/m}{s + b/m}, \qquad K(s) = K_p + s^{-1} K_i$$

Closed-loop sensitivity transfer-functions

$$S(s) = \frac{1}{1 + G(s)\,K(s)} = \frac{s\left(s + \frac{b}{m}\right)}{s^2 + \left(\frac{b}{m} + \frac{p}{m}K_p\right)s + \frac{p}{m}K_i}$$

No complex-poles (oscillations) if

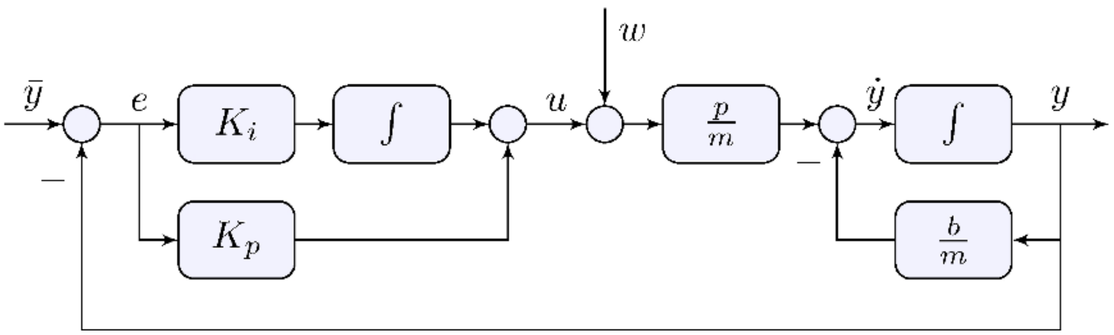
$$0 < 4\frac{p}{m}K_i \leq \left(\frac{b}{m} + \frac{p}{m}K_p\right)^2$$

Example: car model with proportional plus integral feedback

Controller

$$U(s) = \left(K_p + \frac{K_i}{s}\right) E(s) \qquad \Longrightarrow \qquad u(t) = u(0) + K_p e(t) + K_i \int_0^t e(\tau)\, d\tau$$

System + Controller block-diagram



Example: car model with proportional plus integral feedback

Special choice

$$K_i = \frac{b}{m} K_p, \qquad K(s) = K_p + \frac{K_i}{s} = K_p \frac{s + \frac{b}{m}}{s}$$

leads to a pole-zero cancellation

$$G(s)\,K(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}} \times K_p \frac{s + \frac{b}{m}}{s} = \frac{\frac{p}{m} K_p}{s},$$

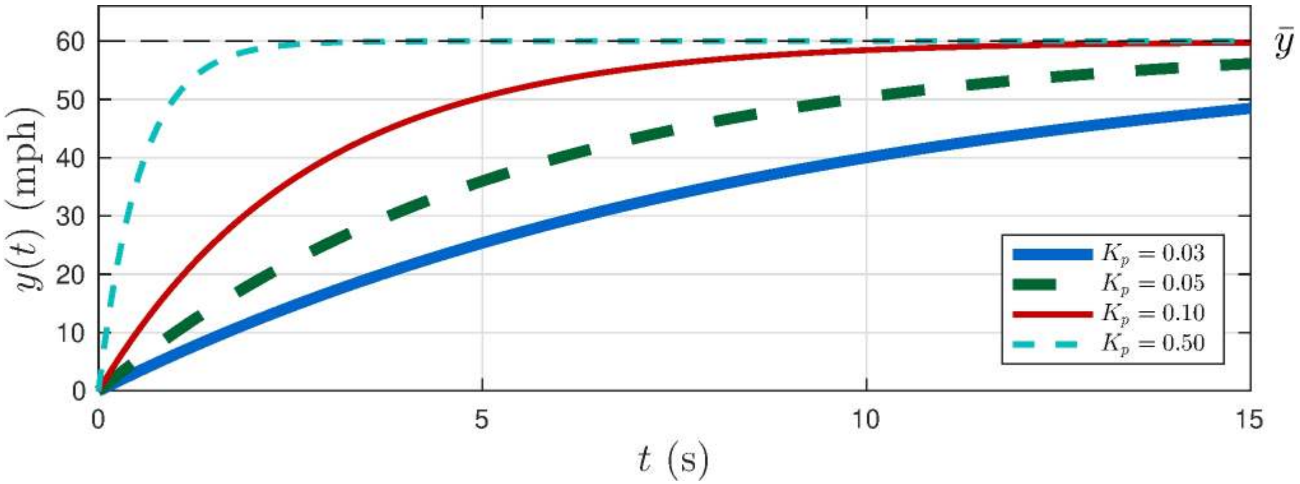
Closed-loop sensitivity transfer-functions

$$S(s) = \frac{1}{1 + G(s)K(s)} = \frac{s}{s + \frac{p}{m} K_p}$$

Zero at the origin: asymptotic tracking

No complex poles: no oscillations

Example: car model with proportional plus integral feedback



Asymptotic tracking, gone are the oscillations!

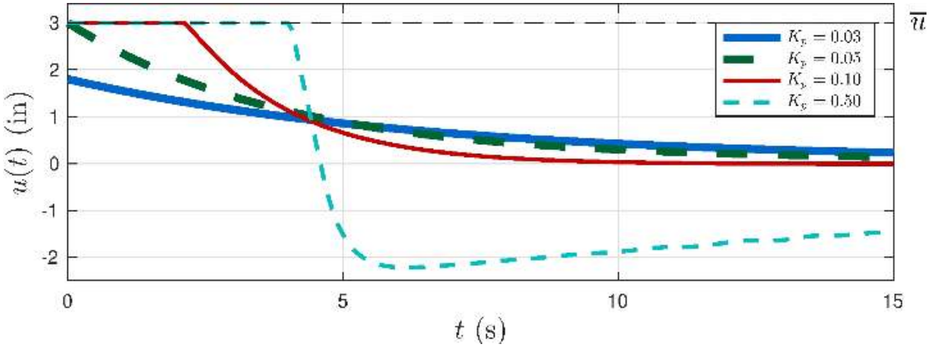
Beware of control effort: saturation occurs if

$$K_p > \frac{\overline{u}}{\bar{y}} = \frac{3}{60} = 0.05$$

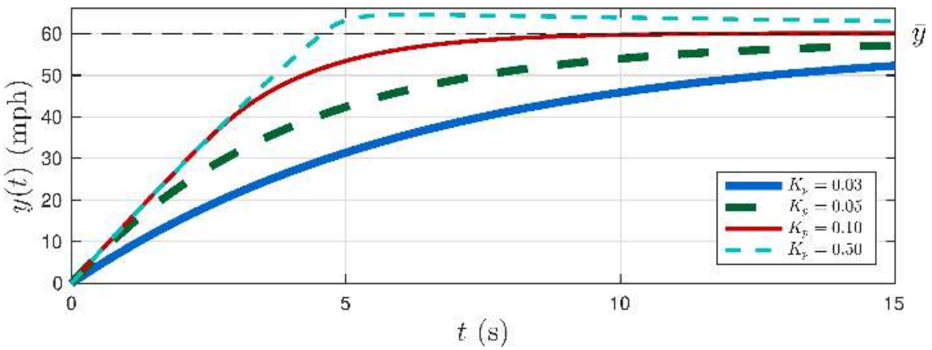
4.3 Integrator Wind-up

Example: car model with proportional plus integral feedback

Nonlinear input saturation



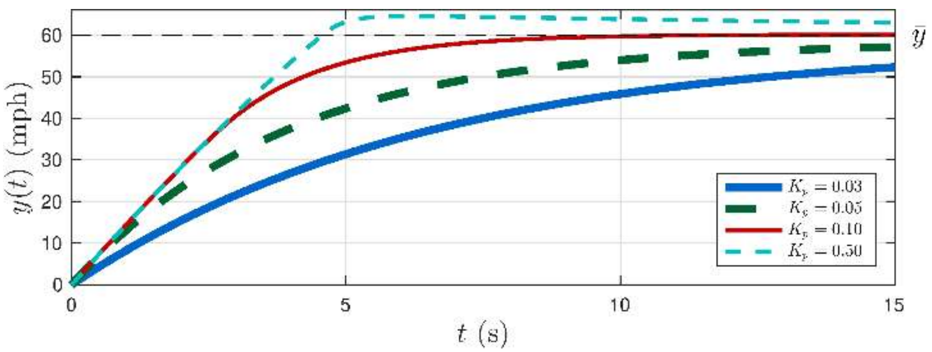
Output overshoots



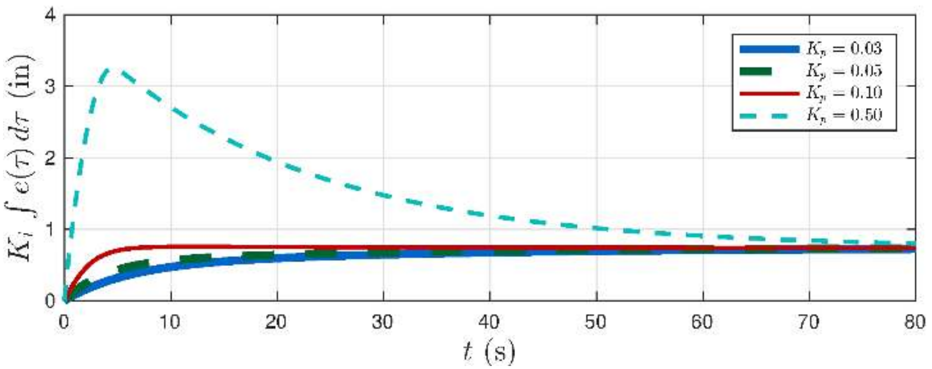
60
50
40
30
20
10
0
$y(t)$ (mph)
0
5
10
15
t (s)
$\bar{y}$
$K_p = 0.03$
$K_p = 0.05$
$K_p = 0.10$
$K_p = 0.50$

Example: car model with proportional plus integral feedback

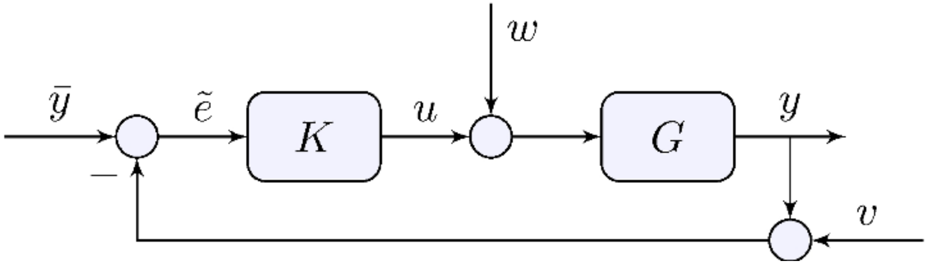
Output converges slowly



Integrator wind-up



4.4 Feedback with Disturbances



Loop relations

$$y = G\,(u + w), \qquad u = K\,\tilde{e}, \qquad \tilde{e} = \bar{y} - (y + v)$$

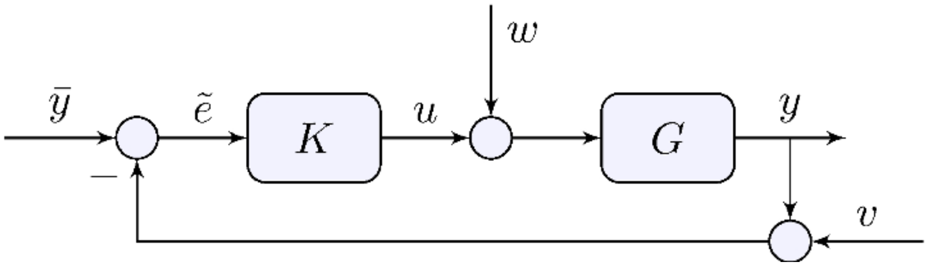
Closed-loop transfer-functions

$$y = H\,\bar{y} - H\,v + D\,w, \qquad u = Q\,\bar{y} - Q\,v - H\,w$$

Tracking error

$$e = \bar{y} - y = S\,\bar{y} + H\,v - D\,w$$

Beware $\tilde{e} \neq e$!

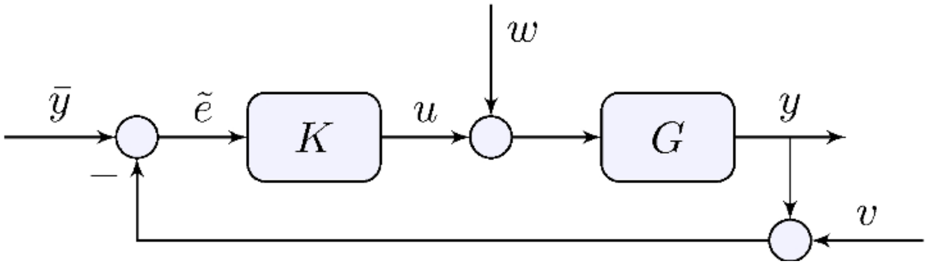


Closed-loop transfer-functions

$$
\begin{aligned}
y &= H\,\bar{y} - H\,v + D\,w \\
u &= Q\,\bar{y} - Q\,v - H\,w \\
e &= S\,\bar{y} + H\,v - D\,w
\end{aligned}
$$

Gang of four

$$
S = \frac{1}{1+GK}, \qquad D = \frac{G}{1+GK} \qquad H = \frac{GK}{1+GK}, \qquad Q = \frac{K}{1+GK}
$$

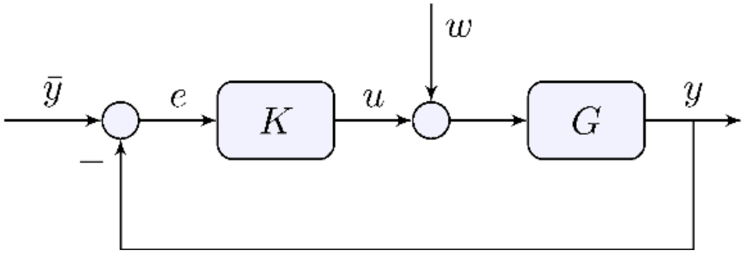


Gang of four

$$S = \frac{1}{1+GK}, \qquad D = GS, \qquad H = GKS, \qquad Q = KS$$

- S has as zeros the poles of GK
- No pole-zero cancellations in $GK \implies S, D, H, Q$ have the same poles!
- Stabilizing S stabilizes all other transfer-functions
- If there are pole-zero cancellations in GK be careful and see Section 4.7

4.5 Input-Disturbance Rejection



Closed-loop transfer-functions

$$e = S\,\bar{y} - D\,w, \qquad D = GS = \frac{G}{1 + GK}$$

Lemma 4.2

Let** $S = (1 + G\,K)^{-1}$ **and** $D = GS$ **be asymptotically stable and

$$\bar{y}(t) = \bar{y}\cos(\omega t + \phi), \qquad w(t) = \bar{w}\cos(\omega t + \psi)$$

***If** K **has a pole at** $s = j\omega$ **then** $\lim_{t\to\infty} e(t) = 0$*

Proof: S and D have zeros at $s = j\omega$

Example: car model with proportional feedback

Closed-loop transfer-functions

$$S(s) = \frac{s + \frac{b}{m}}{s + \frac{b}{m} + \frac{p}{m}K}, \qquad D(s) = S(s)\,G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m} + \frac{p}{m}K}$$

No asymptotic tracking, no asymptotic disturbance rejection

Example: car model with integral feedback

Closed-loop transfer-functions

$$S(s) = \frac{s\left(s + \frac{b}{m}\right)}{s^2 + \frac{b}{m}s + \frac{p}{m}K}, \qquad D(s) = S(s)\,G(s) = \frac{\frac{p}{m}s}{s^2 + \frac{b}{m}s + \frac{p}{m}K}$$

Asymptotic tracking and asymptotic disturbance rejection, oscillations

Example: car model with integral feedback

Closed-loop transfer-functions

$$S(s) = \frac{s\left(s + \frac{b}{m}\right)}{s^2 + \frac{b}{m}s + \frac{p}{m}K}, \qquad D(s) = S(s)\,G(s) = \frac{\frac{p}{m}s}{s^2 + \frac{b}{m}s + \frac{p}{m}K}$$

Asymptotic tracking and asymptotic disturbance rejection, oscillations

Example: car model with proportional plus integral feedback

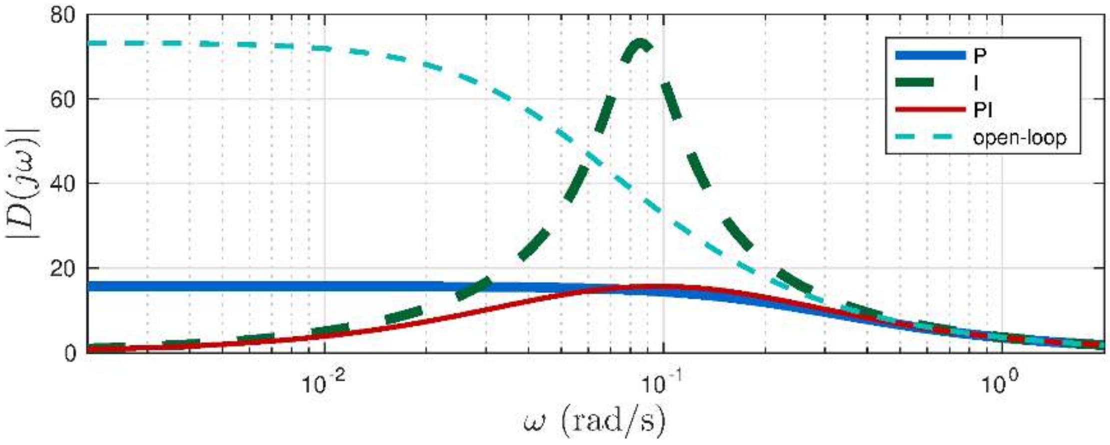
Closed-loop transfer-functions

$$S(s) = \frac{s}{s + \frac{p}{m}K_p}, \qquad D(s) = S(s)\,G(s) = \frac{\frac{p}{m}s}{\left(s + \frac{b}{m}\right)\left(s + \frac{p}{m}K_p\right)}$$

Asymptotic tracking and asymptotic disturbance rejection, no oscillations

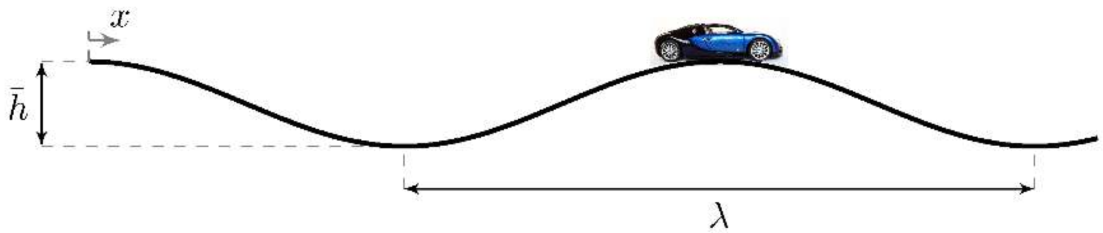
4.5 Input-Disturbance Rejection

Example: car model with various feedback controllers



Integral controller is sensitive to disturbances at 0.1 rad/s

Watch out for rolling hills!



Example: toilet bowl model

Loop relations

$$G(s) = \frac{1}{sA}, \qquad K(s) = K$$

Closed-loop transfer-functions

$$S(s) = \frac{s}{s + \frac{K}{A}}, \qquad D(s) = \frac{\frac{1}{A}}{s + \frac{K}{A}}$$

Asymptotic tracking but no asymptotic disturbance rejection!

Pole must be at the controller for asymptotic disturbance rejection

D does not have a zero at the origin

4.6 Measurement noise

Closed-loop transfer-functions

$$e = S\,\bar{y} + H\,v, \qquad S = \frac{1}{1+GK}, \qquad H = GKS$$

Fundamental limitation

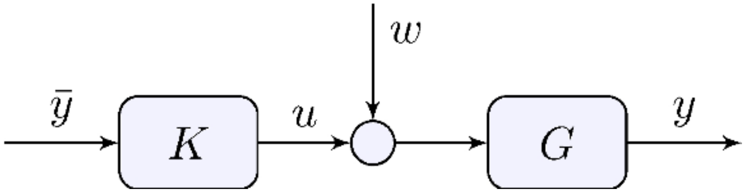
$$S + H = 1$$

At every frequency

$$S(j\omega) + H(j\omega) = 1$$

- Good tracking ($S(j\omega) \approx 0$) means bad measurement noise rejection ($H(j\omega) \approx 1$)
- Good measurement noise rejection ($H(j\omega) \approx 0$) means bad tracking ($S(j\omega) \approx 1$)
- Tracking requires good sensors!
- $|S(j\omega)| + |H(j\omega)| \neq 1$: both S and H can be large at the same time!

4.7 Pole-zero Cancellations and Stability



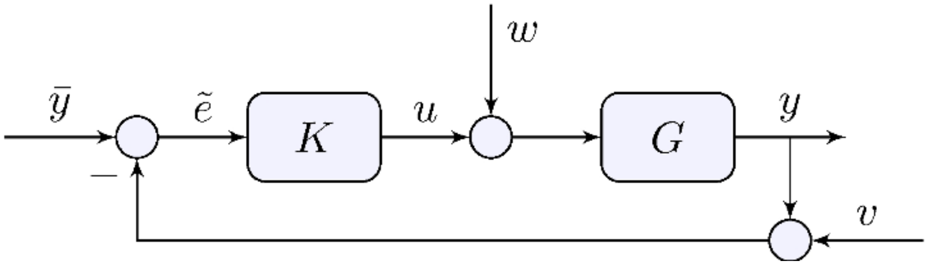
Open-loop

$$G = \frac{N_G}{D_G}, \qquad K = G^{-1} = \frac{D_G}{N_G},$$

- G cannot be strictly proper or K will not be proper
- Even if G is proper but not strictly proper and $K = G^{-1}$

$$y = G\,K\,\bar{y} + Gw = \bar{y} + Gw, \qquad u = K\bar{y}$$

- Both G and K must be stable or y or u will be unbounded
- Stabilization of unstable systems cannot be done in open-loop



Closed-loop

Internal stability if

$$S, \qquad S\,G, \qquad S\,K, \qquad S\,G\,K$$

are all asymptotically stable

Lemma 4.3:

Internal stability if and only if S is asymptotically stable and any pole-zero cancellation in GK is of stable poles or zeros

Proof (sketch): If

$$G = \frac{(s-z)}{(s-p)}\tilde{G}, \qquad K = \frac{(s-p)}{(s-z)}\tilde{K}, \qquad GK = \tilde{G}\tilde{K}, \qquad S = \tilde{S} = \frac{1}{1+\tilde{G}\tilde{K}}$$

However

$$SG = \frac{(s-z)}{(s-p)}\tilde{S}\tilde{G}, \qquad SK = \frac{(s-p)}{(s-z)}\tilde{K}\tilde{G}$$

so that z and p must both have negative real part

Example: car model with proportional plus integral feedback

Model and controller

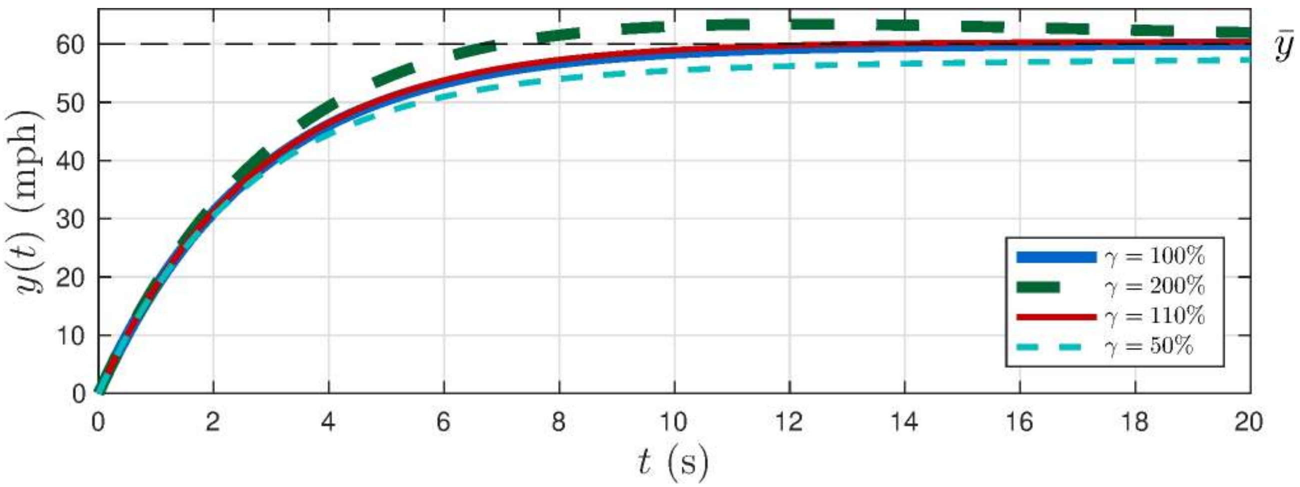
$$G(s) = \frac{p/m}{s + b/m}, \qquad K(s) = K_p \frac{s+z}{s}, \qquad z = \frac{K_i}{K_p} = \frac{b}{m}$$

Closed-loop transfer-functions

$$S(s) = \frac{s}{s + \frac{p}{m}K_p}, \qquad H(s) = \frac{\frac{p}{m}K_p}{s + \frac{p}{m}K_p}, \qquad Q(s) = \frac{K_p\left(s + \frac{b}{m}\right)}{s + \frac{p}{m}K_p}, \qquad D(s) = \frac{\frac{p}{m}s}{\left(s + \frac{b}{m}\right)\left(s + \frac{p}{m}K\right.}$$

Canceled pole is a zero of Q and a pole of D

Example: car model with proportional plus integral feedback



Imperfect cancellation

$$K_i = \gamma \frac{b}{m} K_p$$

More to come in Chapter 6

Learn more

- Book website